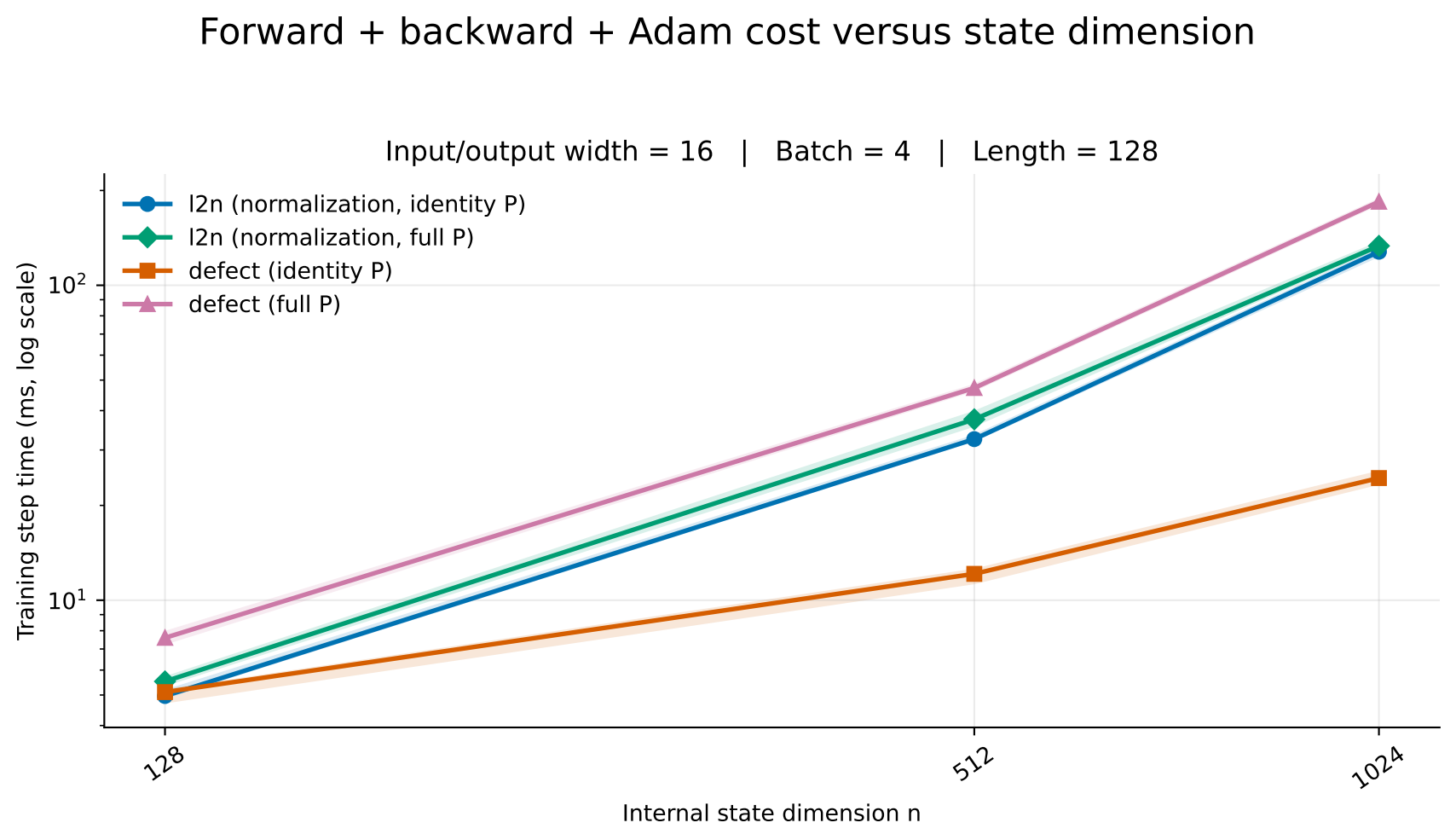
Forward + backward + Adam cost versus state dimension
Input/output width = 16 | Batch = 4 | Length = 128
l2n (normalization, identity P)
l2n (normalization, full P)
defect (identity P)
defect (full P)
Training step time (ms, log scale)
10^2
10^1
128
512
1024
Internal state dimension n
CPU · float32 · 1 CPU thread(s) · eager · l2n: scan, l2n(I) basis: learned · defect: scan · Adam lr=0.001
Median of 20 repetitions; band = 25th–75th percentile; 5 warmups.
Includes parameter construction, gradient reset, forward, loss, backward, and Adam updates; excludes optimizer initialization.
Equal dimensions and inputs; full/full shares the generic LTI family; initial transfer maps differ.